

$H \rightarrow \gamma\gamma$: Cross-Section Measurement Using the ATLAS Detector

Third Year Lab Report

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This report presents a measurement of the Higgs boson production cross-section in the diphoton channel using ATLAS data collected at a centre-of-mass energy of 13 TeV. Selection cuts were applied primarily on photon momentum, isolation, and identification variables. The signal in the 120–130 GeV region was modelled using a triple-Gaussian model, while the background was described by various functional forms fitted to sideband regions. The Higgs contribution was extracted by fitting the diphoton invariant-mass distribution with the combined signal and background model. Variations in binning, sideband choices, selection cuts, and all models were used to evaluate statistical and systematic uncertainties. The resulting cross-section is

$$\sigma(H \rightarrow \gamma\gamma) = 65.9 \pm 7.67_{\text{syst}} \pm 10.4_{\text{stat}} \pm 1.38_{\text{lumi}} \text{ fb.}$$

1. INTRODUCTION

The Higgs boson plays a central role in the Standard Model as the particle responsible for giving mass to quarks, leptons, and the weak gauge bosons [1]. Following its discovery at the LHC, the focus shifted to precision studies, since deviations in its production or decay rates could signal physics beyond the Standard Model. Among its decay channels, the diphoton mode is especially powerful: despite its small branching ratio, it offers a clean signature with precisely measured photon energies, producing a sharp 125 GeV resonance well suited for detailed Higgs characterisation. The Higgs boson in this study is produced in 13 TeV proton–proton collisions [2], primarily through gluon–gluon fusion and vector-boson fusion, and decays into two photons. These events occur in a dense collision environment in ATLAS, where numerous other processes produce real photons and where jets can be misidentified as photons, leading to a background that dominates the diphoton invariant-mass spectrum. Only a narrow mass window contains the genuine Higgs contribution. To isolate it, the analysis used diphoton kinematic and identification variables to suppress background, and then fitted signal and background models to the observed invariant-mass distribution. This procedure enabled the Higgs contribution to be extracted and its production cross-section to be determined.

2. VARIABLES AND EQUATIONS

2.1. Event Variables

In this analysis, photon-related event variables were used to describe their kinematics, identification, and isolation. The transverse momentum p_T measures the photon’s momentum perpendicular to the beam axis. The pseudorapidity η describes its direction relative to the beam axis, while the azimuthal angle ϕ specifies its orientation around the beam axis. The energy E is reconstructed from the electromagnetic calorimeter. Photon isolation is quantified using track-based and calorimeter-based measures. Track isolation is the sum of transverse momenta of tracks within a cone of radius $\Delta R = 0.2$ around the photon. Topocluster isolation measures the sum of calorimeter energy deposits within a larger cone of radius $\Delta R = 0.4$, pro-

viding a complementary assessment of the photon’s isolation.

Photon identification is described by loose ID and tight ID, based on calorimeter shower shapes, with the tight selection yielding higher purity. Similarly, loose isolation and tight isolation indicate whether the photon satisfies the corresponding isolation criteria. The number of photons in the event was also recorded, as well as whether the event fired the relevant photon trigger. These variables together provided precise event selections.

2.2. Cross-Section and Significance Equations

The cross-section σ quantifies the probability for a given process to occur in proton–proton collisions. It represents the effective interaction area for producing the final-state photons. In this analysis, the measured cross-section was obtained from the background-subtracted event yield using

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\varepsilon \int L dt}, \quad (1)$$

where N_{sel} is the number of selected events, N_{bkg} is the background estimate, $\int L dt$ is the integrated luminosity of the dataset, and ε is the signal efficiency [3].

To assess the strength of the observed signal relative to statistical fluctuations in the data, the analysis used the significance

$$S = \frac{\sum w_{\text{signal}}}{\sqrt{\sum w_{\text{data}}}}, \quad (2)$$

where $\sum w_{\text{signal}}$ is the total signal weight after the event selection, and $\sum w_{\text{data}}$ is the total event weight in data. This provides a direct quantitative measure of how prominently the signal emerges above the statistical background.

3. SELECTION CUTS AND MODELS

3.1. Selection Cuts

All selection cuts were chosen through the significance optimisation based on Equation 2. Events were required to contain exactly two photons, capturing 98% of the events. Photons in the calorimeter transition

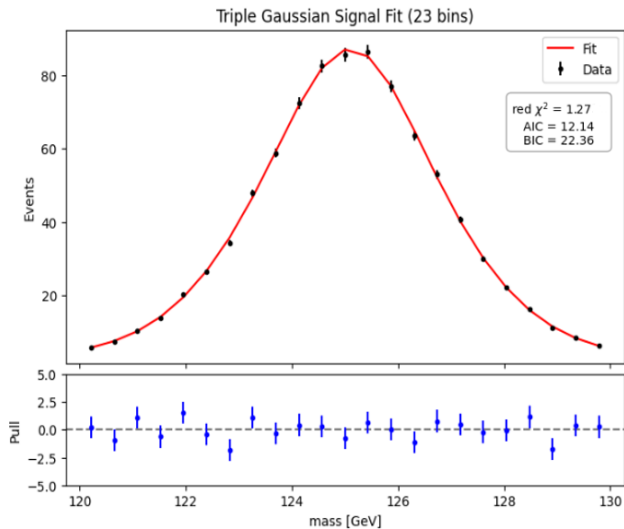


FIG. 1. Triple-Gaussian fit to the diphoton invariant-mass distribution in the 120–130 GeV range (23 bins). The lower panel shows fit residuals, and the annotation box reports fit-quality metrics

regions $1.37 < |\eta| < 1.52$ were excluded due to unreliable energy reconstruction [4].

Identification and isolation variables were used because they gave better significance performance than cone-based options. Four selection configurations were tested. Three used different combinations of identification and isolation working points (tight-loose, loose-tight, and tight-tight) to probe the efficiency-purity balance. The standard photon requirement was $p_T > 30$ GeV, while the fourth configuration applied a higher p_T cut motivated by the significance optimisation.

3.2. Signal and Background Models

The 120–130 GeV window was chosen after testing several intervals and selecting the one with the most stable χ^2 performance, and it was used to model the diphoton signal. Three functional forms were evaluated within this window: a single Gaussian, a double Gaussian, and a triple Gaussian. Each was fitted to the simulated diphoton signal, and the fit quality was assessed using the reduced χ^2 and overall fit stability. Across all cut configurations, the triple-Gaussian model showed the most stable behaviour and provided the most reliable description of the signal, and was therefore adopted for the analysis. An example of the resulting fit is shown in Figure 1.

To model the diphoton background, five candidate functions were initially explored: a seventh-order polynomial, a three-parameter exponential, a four-parameter dijet function, a seventh-order Laurent series, and a rational function of two over two. The exponential form was discarded due to poor consistency and significantly worse reduced χ^2 performance. The remaining four models were used in the cross-section analysis.

To avoid bias from the Higgs signal peak, the 120–130 GeV region was masked during the background fit, ensuring that events in this window did not contribute to the determination of the background parameters. An example background fit is shown in Figure 2.

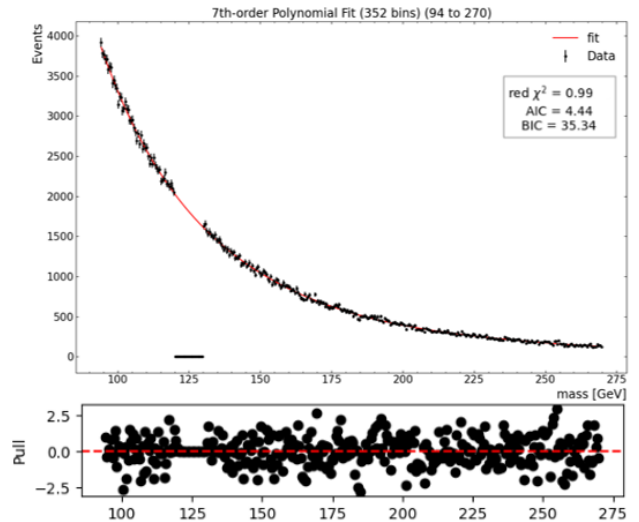


FIG. 2. seventh-order polynomial fit to the diphoton invariant-mass distribution in the 94–270 GeV range (0.5 GeV bins). Lower panel shows residuals; annotation box reports fit-quality metrics.

4. CROSS-SECTION MEASUREMENT

4.1. Signal Extraction

To extract the signal yield in data, the triple-Gaussian signal model was combined with each of the four selected background models. The Gaussian means and widths were fixed to those obtained from the signal-only fit shown in Figure 1. The amplitude of the narrowest Gaussian was left free, while the amplitudes of the two wider Gaussians were constrained to fixed ratios relative to the narrow component, preserving the shape determined from the simulation.

Each combined model was then fitted to the diphoton mass spectrum using the same binning and mass range as in the background-only fit shown in Figure 2, ensuring consistency across all background models. Since the signal-only fit used a different bin width, the extracted amplitude was scaled by the ratio of the bin widths in the signal-only and background fits.

This procedure yielded the excess events in the signal window for each background model. An example of the combined triple-Gaussian and background model fit, now including the 120–130 GeV signal region, is shown in Figure 3.

4.2. Cross-Section Calculation

In order to calculate the $H \rightarrow \gamma\gamma$ cross-section, the amplitude extracted from the signal-extraction procedure was inserted into Equation 1. The amplitude provided the numerator of the expression, while the efficiency was calculated from 70% of the signal sample that had been used in the analysis. Because the same fraction of the dataset was analysed, the integrated luminosity of 36.6 fb^{-1} was correspondingly reduced, yielding the effective luminosity associated with the subset of data. From the amplitude, the efficiency, and the effective luminosity, the cross-section value was obtained.

4.3. Uncertainties and Final Result

Using a wide range of background-fit configurations by varying both the fit ranges and the binning, together with multiple binning choices for the signal model,

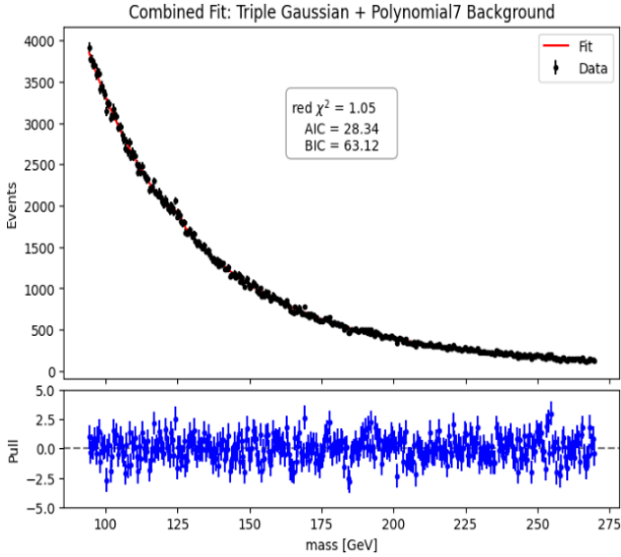


FIG. 3. Combined triple-Gaussian signal and seventh-order polynomial background fit to the diphoton invariant-mass distribution in the 94–270 GeV range (0.5 GeV bins). Lower panel shows residuals; annotation box reports fit-quality metrics.

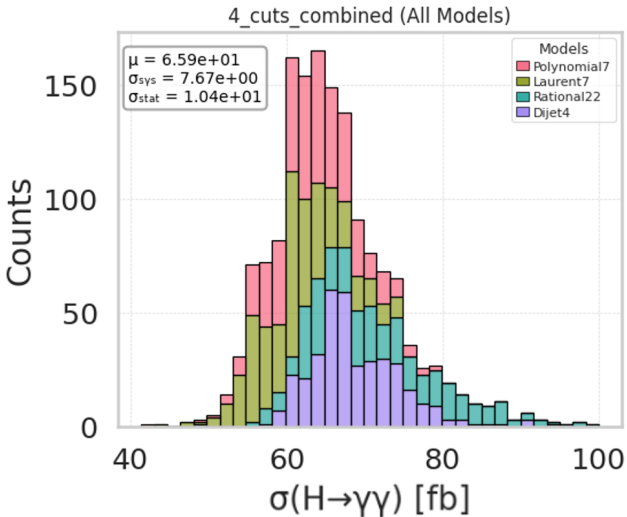


FIG. 4. Combined cross-section distribution for the four selection cuts with models shown. The annotation box displays the mean value and uncertainties.

and repeating this procedure for every selection configuration defined in Section 3.1, we obtained around 1600 independent cross-section values across all models. The cross-section distribution is shown in Figure 4. The final quoted cross-section was taken as the

mean of this distribution.

The systematic uncertainty was defined as the standard deviation of this distribution. To keep the models under control and avoid contributions from poorly behaved fits, reduced χ^2 constraints were applied so that only statistically reliable fits were included when forming the distribution.

For the statistical uncertainty, the cross-section values entering the main distribution were computed using 70% of the dataset. To obtain a more robust estimate of the statistical uncertainty itself, we used 100 percent of the data. For each of the four selection cuts, 20 cross-section values were generated using the full dataset, and the value closest to the mean of that cut’s cross-section distribution was identified. The statistical error associated with that value was assigned as the statistical uncertainty for that cut. This produced four statistical-uncertainty estimates, one for each cut, and the RMS of these four values was taken as the statistical uncertainty of the final combined distribution shown in Figure 4.

For the luminosity uncertainty, we referred to the 2019 ATLAS CONF documentation [5], which reports a luminosity uncertainty of 2.1%. For the integrated luminosity used in this analysis, this corresponds to a 1.38 fb luminosity uncertainty.

The final measured cross-section is

$$\sigma(H \rightarrow \gamma\gamma) = 65.9 \pm 7.67_{\text{syst}} \pm 10.4_{\text{stat}} \pm 1.38_{\text{lumi}} \text{ fb}.$$

5. CONCLUSION

The cross-section obtained is consistent with expectations. The 2023 ATLAS result at 13.6 TeV reported

$$76^{+9}_{-7\text{syst}} \pm 11_{\text{stat}} \text{ fb},$$

[6] since this analysis uses 13 TeV data, a lower cross-section is expected. The Standard Model prediction for the $H \rightarrow \gamma\gamma$ cross-section at 13 TeV is 55.6 ± 2.5 fb [7], which is consistent with our measurement within statistical uncertainties, as expected given the smaller dataset and the simplified analysis strategy used.

There are several ways the experiment could have achieved improved precision. Using a Crystal Ball signal model [8], which is standard in particle physics analyses, would better capture the non-Gaussian tail structure of the diphoton mass peak and reduce shape-related systematic uncertainty. Background modelling could be strengthened by incorporating functions from different functional families to reduce bias, while also limiting the models to those that are physically motivated; for example, the seventh-order polynomial showed unphysical oscillations and should be excluded.

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